8 Damping Power System Oscillations Using Grid-Forming Converters

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8.1 INTRODUCTION

Nowadays, it is a well-known fact that the high penetration of renewable power plants has completely changed the electrical generation model. In some power systems, renewable energy sources (RES), based on power electronic converters (PECs), are displacing conventional synchronous generators (SGs) reducing system inertia and fault current contribution [1]. Although RES are increasing their contribution to grid ancillary services, in order to maintain a safe and reliable power systems operation, some services, such as power system restoration or low-frequency oscillation (LFO) damping, are requested only to SGs. Nevertheless, new grid-forming converters (GFCs), emulating SGs behavior, can contribute to provide these services as well [2–8].

This chapter proposes two power system stabilizers (PSSs) for damping LFO in power systems using PECs. First, a power stabilizer, implemented in a grid-forming converter, has been used to provide LFO damping by acting on the converter power

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angle and exchanging active power (POD-P). On the other hand, a second power stabilizer provides LFO damping through the reactive power (POD-Q) when the power converter is used as a STATCOM using the voltage control loop to damp power oscillations. Both power stabilizers have been compared in two case studies. In the first case, LFO damping in a single machine connected to an infinite bus through a tie-line is analyzed, and in the second case, damping of inter-area oscillation modes, in a two-area system, is also studied. Simulation results and stability studies demonstrate that both stabilizers damp power oscillations, being the POD-P stabilizer more effective than the POD-Q, although the former requires some kind of energy provision in the DC bus. System description and modeling of an SG and a PEC, both connected in parallel to the same bus, are shown in the following section. Next, control schemes of the grid-forming POD-P and STATCOM POD-Q stabilizers are proposed. From these two stabilizers, a small-signal analysis is carried out, studying the influence of both stabilizers in the generator's electromechanical and flux-decay modes. This section ends with a time-domain simulation comparing the damping of LFO modes in each case.

Subsequently, the damping performance of both stabilizers is checked in a two-area system. Both areas are interconnected through a transmission line. The dynamic response of this system is analyzed under a load change in one area and during a line tripping. As in the previous case, an eigenvalues analysis is also carried out when the reactance of the interconnection line changes, with and without POD compensation.

8.2 SYSTEM DESCRIPTION AND MODELING

Figure 8.1 shows the connection of an SG and a PEC to an infinite bus through a reactance X_e with the purpose of damping power oscillations. Both devices are connected in parallel to the same bus where the voltage vector is represented by $V_t e^{j\theta}$, being V_t the voltage magnitude and θ its angle with respect to infinite bus $V_\infty \angle 0°$. The current vector through the reactance X_e is $I_G e^{j\gamma}$.

In LFO studies, it is common to use SGs reduced-order models. Here, a simple one-axis (flux-decay) model has been considered. This is a four-order model [9] using E'_q as a state variable that represents the field flux dynamics but oriented to the quadrature axis, q. The position of this q-axis with respect to the reference voltage $V_\infty \angle 0°$ is the angle δ that represents the second variable state. The dynamic

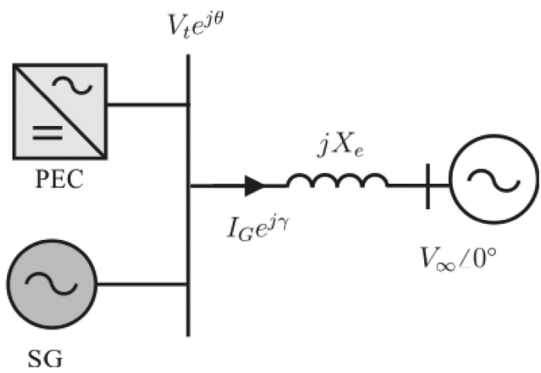


FIGURE 8.1 SG and PEC in a single-machine infinite-bus system.

model is completed by defining the generator speed ω and the excitation voltage E_{fd} as the other state variables. In this model, faster dynamics, including stator/network dynamics, have been neglected. From a complete set of nonlinear SG dynamic equations, a state-space model is obtained by linearizing them at a given operating point. The state matrix eigenvalues determine the oscillation modes that can be damped using suitable power stabilizers. The dynamic equations of the model are shown below.

The dynamic equation of E'_q can be expressed as

$$T'_{d0}\frac{dE'_q}{dt} = -E'_q - \left(X_d - X'_d\right)I_d + E_{fd} \tag{8.1}$$

where T'_{d0} is the d-axis open-circuit transient time constant. The reactance X_d and X'_d are the total and transient d-axis reactance, respectively. The equation of motion of an SG is as follows:

$$2H\frac{d\bar{\omega}}{dt} = T_m - T_e - D\left(\bar{\omega} - \bar{\omega}_0\right) \tag{8.2}$$

where H is the inertia constant of the generator, expressed in seconds, D is the damping constant, and $\bar{\omega}$ is the angular velocity of the rotor, and $\bar{\omega}_0$ is the reference angular speed, both magnitudes expressed in pu. Meanwhile, T_m is an input of the model and represents the mechanical torque also in pu. T_e is the electrical torque, which is expressed as

$$T_e = E'_q I_q + \left(X_q - X'_d\right)I_d I_q \tag{8.3}$$

Note that the electrical torque is calculated as a nonlinear expression of electrical variables and the SG parameters. X_q is the total q-axis reactance, and I_d and I_q are the dq-components of the current vector. The dynamic equation of δ is obtained by integrating the rotational speed of the generator, ω, with respect to a synchronously rotating reference, ω_s, in rad/s. In pu, the dynamic equation of δ is as follows

$$\frac{1}{\omega_s}\frac{d\delta}{dt} = \bar{\omega} - \bar{\omega}_0 \tag{8.4}$$

where $\bar{\omega}$ is the speed expressed in pu and $\bar{\omega}_0 = 1$ pu. Assuming a fast exciter for regulating voltage V_t at the output terminal from the reference voltage V_{ref}, the dynamic equation of E_{fd} is equal to

$$T_A\frac{dE_{fd}}{dt} = -E_{fd} + K_A\left(V_{ref} - V_t\right) \tag{8.5}$$

where T_A and K_A are the time constant and the gain of the exciter.

Figure 8.2 shows the vector diagram of the voltage and current of the SG. The q-axis of the generator forms an angle δ with respect to a D-axis (aligned to the

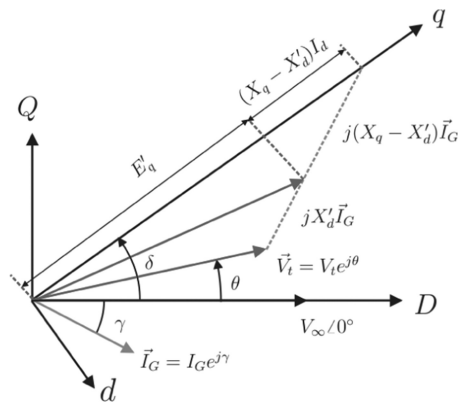


FIGURE 8.2 SG voltage and current diagram.

reference voltage vector $V_\infty \angle 0°$ at the infinite bus). The angle θ corresponds to the voltage $\vec{V}_t = V_t e^{j\theta}$ at the generator terminals.

Next expressions complete the set of algebraic equations

$$X_q I_q - V_d = 0 \tag{8.6}$$

$$E'_q - V_q - X'_d I_d = 0 \tag{8.7}$$

Figure 8.3 shows an equivalent circuit of the SG and the PEC corresponding to the single-machine infinite-bus system of Figure 8.1. The SG is represented as a dependent voltage source $\vec{E}_{SG}$ [9]

$$\vec{E}_{SG} = \left[\left(X_q - X'_d\right) I_q + jE'_q\right] e^{j\left(\delta - \frac{\pi}{2}\right)} \tag{8.8}$$

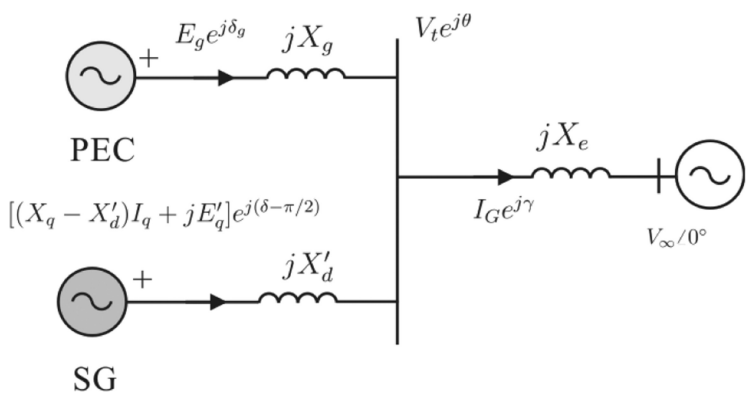


FIGURE 8.3 Equivalent circuit of a one-axis SG and a PEC.

behind the reactance X'_d. Similarly, the PEC is modeled as an independent voltage source $E_g e^{j\delta_g}$ behind a reactance X_g.

The voltage vector $V_t e^{j\theta}$, shown in Figure 8.3, is obtained by applying Millman's theorem in terms of $E_g e^{j\delta_g}$ and $V_\infty \angle 0°$ as follows:

$$V_t e^{j\theta} = \left(V_d + jV_q\right) e^{j\left(\delta - \frac{\pi}{2}\right)} = +K'_d \left[\left(X_q - X'_d\right) I_q + jE'_q\right] e^{j\left(\delta - \frac{\pi}{2}\right)} + K_g jE_g e^{j\left(\delta_g - \frac{\pi}{2}\right)} + K_e V_\infty \tag{8.9}$$

where K'_d, K_g, and K_e are

$$K'_d = \left(\frac{\overline{Y}'_d}{\overline{Y}'_d + \overline{Y}_g + \overline{Y}_e}\right) = \frac{X_g X_e}{X_g X_e + X'_d X_e + X'_d X_g} \tag{8.10}$$

$$K_g = \left(\frac{\overline{Y}_g}{\overline{Y}'_d + \overline{Y}_g + \overline{Y}_e}\right) = \frac{X'_d X_e}{X_g X_e + X'_d X_e + X'_d X_g} \tag{8.11}$$

$$K_e = \left(\frac{\overline{Y}_e}{\overline{Y}'_d + \overline{Y}_g + \overline{Y}_e}\right) = \frac{X'_d X_g}{X_g X_e + X'_d X_e + X'_d X_g} \tag{8.12}$$

being the admittances $\overline{Y}'_d = -j/X'_d$, $\overline{Y}_g = -j/X_g$ and $\overline{Y}_e = -j/X_e$. As these admittances have only imaginary part K'_d, K_g and K_e appear as constants in equation (8.9).

By multiplying equation (8.9) by $e^{-j\left(\delta - \frac{\pi}{2}\right)}$ and separating into real and imaginary parts, the following algebraic equations of the network are obtained:

$$V_d - K'_d \left(X_q - X'_d\right) I_q + K_g E_g \sin\left(\delta_g - \delta\right) - K_e V_\infty \sin\delta = 0 \tag{8.13}$$

$$V_q - K'_d E'_q - K_g E_g \cos\left(\delta_g - \delta\right) - K_e V_\infty \cos\delta = 0 \tag{8.14}$$

PEC's active and reactive power expressions, expressed in pu, are

$$P_g = \left(\frac{V_t}{X_g}\right) E_g \sin\left(\delta_g - \theta\right) \tag{8.15}$$

$$Q_g = \left(\frac{V_t}{X_g}\right) \left[E_g \cos\left(\delta_g - \theta\right) - V_t\right] \tag{8.16}$$

According to equations (8.15) and (8.16), it is well known that the active power P_g is controlled acting on the angle δ_g and the reactive power Q_g by the voltage control of E_g.

PEC control is designed to damp power oscillations by exchanging active and reactive power, acting on the angle and module of its internal voltage. In this chapter, two stabilizers are compared, one implemented in a grid-forming converter and the other in a STATCOM. The fundamentals of both stabilizers are presented below.

8.3 GRID-FORMING CONVERTER STABILIZER

Unlike conventional grid-following (GFL) converters, grid-forming converters operate as a real voltage source maintaining an internal voltage phasor that enables to immediate response to changes in the external system without using any phase-locked loop (PLL) to synchronize it to the grid. Instead of a PLL, a grid-forming converter uses a synchronization loop based on the active power response as SGs do.

Figure 8.4 shows the synchronization loop where the power angle δ_g is obtained from the difference of active power reference P_g^* and the actual power P_g. As previously mentioned, δ_g defines the angular position of the internal voltage vector $E_g e^{j\delta_g}$ with respect to reference voltage $V_\infty \angle 0°$. According to equation (8.15), the active power P_g depends on δ_g and the voltage vector $V_t e^{j\theta}$. The derivative of angle δ_g is obtained as the difference between the internal angular speed ω_g and the angular speed corresponding to the reference frequency ω_s.

Considering that $\bar{\omega}_g$ is expressed in pu and $\bar{\omega}_0 = 1$ pu, the power angle δ_g is calculated as

$$\frac{1}{\omega_s}\frac{d\delta_g}{dt} = \bar{\omega}_g - \bar{\omega}_0 \tag{8.17}$$

This equation is completely analogous to equation (8.4) that defined the SG.

As shown in Figure 8.4, the internal frequency $\bar{\omega}_g$ is obtained in terms of P_g^* and P_g as

$$\bar{\omega}_g = \bar{\omega}_0 + \frac{P_g^* - P_g}{J_g s + D_g} \tag{8.18}$$

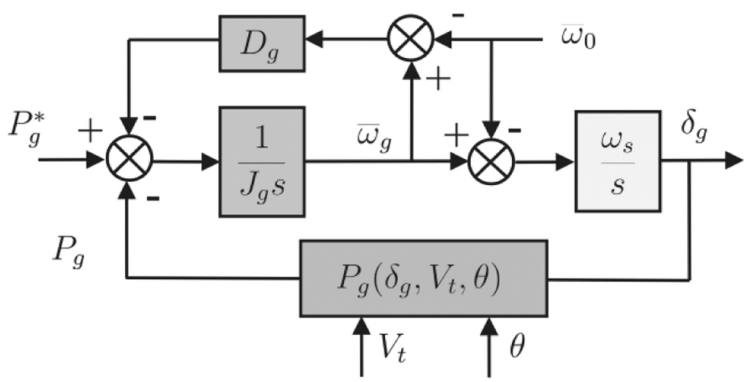


FIGURE 8.4 Power synchronization loop of a grid-forming converter.

where J_g is the inertia moment of the grid-forming (in seconds), and D_g is denoted as a damping constant. The inverse of the damping constant $R_g = 1/D_g$ is known as the droop constant that, in pu, is the ratio of frequency variation in pu, $\Delta f/f_n$, and the normalized power deviation $\Delta P/P_n$. When the inertia constant is equal to zero, $\overline{\omega}_g$ is calculated as the product of a droop constant and the active power increment with respect to a reference, $\overline{\omega}_g - \overline{\omega}_0 = R_g\left(P_g^* - P_g\right)$. As shown in Figure 8.5, in a grid-forming converter, LFO damping is achieved by acting on the internal frequency reference of the power synchronization loop. The signal $\overline{\omega}'$, that is subtracted to the internal frequency $\overline{\omega}_g$, is obtained from the measured frequency ω at the output terminals of the grid-forming converter through a washout function as follows:

$$\overline{\omega}' = K_w\left(\frac{T_w s}{T_w s + 1}\right)\overline{\omega} \tag{8.19}$$

In an SG, its PSS uses a function similar to equation (8.19), but here a phase compensation block is not needed, because in an SG, a lead-lag function is needed to compensate the delay of the exciter effect on the generator torque. However, in a grid-forming converter, it is not required since an immediate action on the active power is obtained when the internal frequency $\overline{\omega}_g$ is modified. The derivative term in the washout filter expressed in equation (8.19) blocks any DC component existing

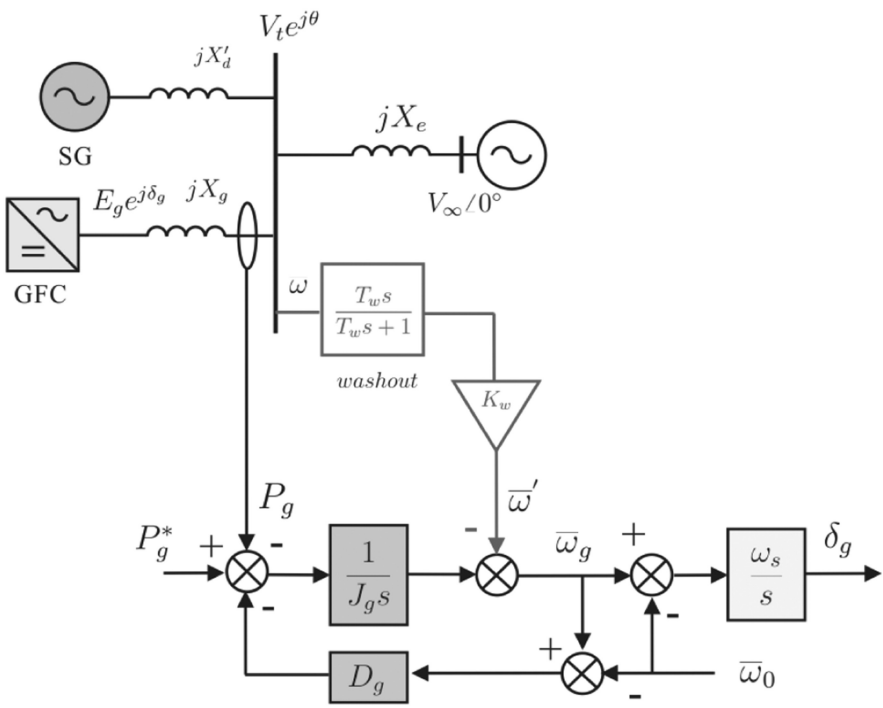


FIGURE 8.5 Block diagram of a grid-forming stabilizer.

on $\bar{\omega}$ as a consequence of a permanent deviation from the rated frequency ω_s. Furthermore, the time constant T_w is chosen in a range of 0.1–20 s so that LFOs measured in the bus can pass unaltered. So, this stabilizer operates only in the range of low-frequency oscillations, while the gain K_w is tuned in order to provide enough damping to the oscillation modes.

The proposed stabilizer acts directly over δ_g and therefore on the active power. Note that a pole-zero cancelation occurs between the stabilizer filter and the integral function ω_s/s. In a GFL converter, a stabilizer of this type can be implemented by calculating the power reference as a function of the frequency measured on the bus. However, its effectiveness is lower due to the dynamics between P_g^* and P_g. As mentioned above, here, the stabilizer acts on angle δ_g, keeping voltage magnitude E_g constant, and also if $E_g = V_t$, according to equation (8.16), the reactive power Q_g is approximately zero, since the angular difference between δ_g and θ is usually small. In the linearized grid-forming model, in order to simplify the study, a zero-inertia has been considered.

Considering this assumption, the frequency deviation $\Delta\omega_g$ is equal to

$$\Delta\omega_g = -R_g \Delta P_g = -R_g K_s \left(\Delta\delta_g - \Delta\theta\right) \tag{8.20}$$

where ΔP_g is obtained by linearizing equation (8.15) when E_g is kept constant, being the synchronization constant K_s

$$K_s = \left(\frac{V_t^0 E_g^0}{X_g}\right)\cos\left(\delta_g^0 - \theta^0\right) \tag{8.21}$$

Assuming that $X_g = 0.15$ pu, $V_t^0 = E_g^0 = 1$ pu, and $\delta_g^0 = \theta^0$, the synchronization constant value is $K_s = 6.67$ pu. According to the values given in Table 8.3 (Annex A), K_s is significantly higher than the synchronization constant on an SG, which means that the active power transmission is achieved with lower δ_g angles.

Figure 8.6 shows a block diagram of the grid-forming stabilizer represented in equation (8.19). The output from this block is the frequency compensation variable $\Delta\bar{\omega}'$, that is proportional to the difference between $\Delta\bar{\omega}$ and Δz, as

$$\Delta\bar{\omega}' = K_w\left(\Delta\bar{\omega} - \Delta z\right) \tag{8.22}$$

where Δz is an auxiliary variable state, and K_w is the stabilizer gain.

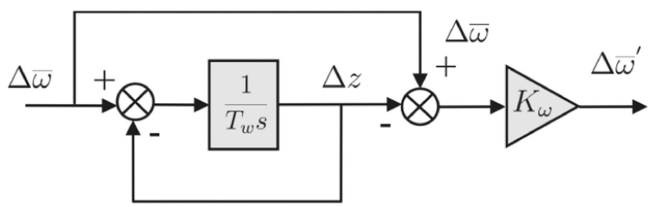


FIGURE 8.6 Block diagram of the washout filter.

A linearized block diagram of the power synchronization loop for a grid-forming converter is shown in Figure 8.7. This figure includes the POD-P active power stabilizer block. Furthermore, it is considered that the linearized active power reference ΔP_g^* is zero, and according to equation (8.20), the actual active power ΔP_g is proportional to the difference of angles $\Delta\delta_g$ and $\Delta\theta$.

By linearizing the expression $V_d = V_t \sin(\delta - \theta)$, the incremental variable $\Delta\theta$ is obtained in terms of $\Delta\delta$ and the algebraic variables ΔV_d and ΔV_q as follows

$$\Delta\theta = \Delta\delta + \left(\frac{E_g^0}{X_g K_s}\right)\left[\Delta V_t \sin\left(\delta^0 - \theta^0\right) - \Delta V_d\right] \tag{8.23}$$

where ΔV_t is expressed as a function of ΔV_d and ΔV_q according to

$$\Delta V_t = \left(\frac{V_d^0}{V_t^0}\right)\Delta V_d + \left(\frac{V_q^0}{V_t^0}\right)\Delta V_q \tag{8.24}$$

The GFC connection increases the number of dynamic equations by two. Linearizing equation (8.17) and considering equation (8.20), the dynamic equation of $\Delta\delta_g$ is obtained as

$$\frac{1}{\omega_s}\frac{d\Delta\delta_g}{dt} = -R_g K_s\left(\Delta\delta_g - \Delta\theta\right) - K_w\left(\Delta\overline{\omega} - \Delta z\right) \tag{8.25}$$

On the other hand, a second dynamic equation corresponding to the auxiliary variable Δz is expressed as

$$T_w\frac{d\Delta z}{dt} = \Delta\overline{\omega} - \Delta z \tag{8.26}$$

Thus, the complete set of dynamic equations correspond to a sixth-order model, which consists of the grid-forming equations (8.25) and (8.26) and the four SG equations (8.1), (8.2), (8.4), and (8.5). While the algebraic equations of the system, that

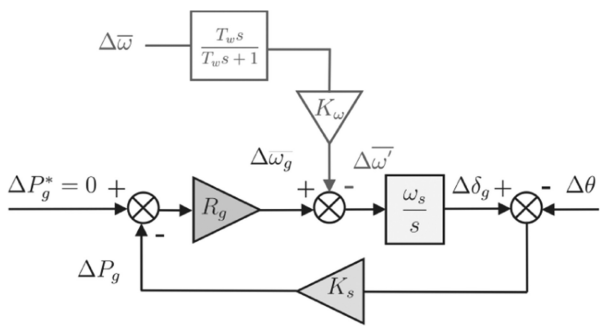


FIGURE 8.7 Linearized block diagram. Grid-forming converter with a POD-P stabilizer.

complete the model, include equations (8.6) and (8.7), which correspond to the SG, and another two that are obtained by linearizing equations (8.13) and (8.14) considering that E_g is constant. Finally, matrices A, B, C, and D of the linearized model are obtained using this set of differential and algebraic equations. The analytical expression of these matrices is given in Appendix B.

8.4 STATCOM STABILIZER

Power system oscillations damping through a STATCOM is carried out acting on the voltage magnitude similarly to conventional PSS of SGs, but without using any phase compensator. In the case of SGs, a PSS modifies the active power generated acting on the voltage E_g (equation 8.15) through the exciter and damping the generator rotor angle swings. A STATCOM can produce the same effect by modifying the bus voltage through the injection of reactive power.

STATCOM models, in transient stability studies, are usually represented by some algebraic equations and the dynamics of the DC voltage. For the sake of simplicity, in this analysis, the dynamic of the DC voltage has not been considered so that the STATCOM only exchanges reactive power in such a way that $\delta_g = \theta$. According to equation (8.15), the active power is proportional to $sin\left(\delta_g - \theta\right)$ so that this active power is zero when both angles are the same. LFO damping is now achieved by changing the voltage reference E_g^* through the signal E_g' which is the output of a washout filter shown in Figure 8.8.

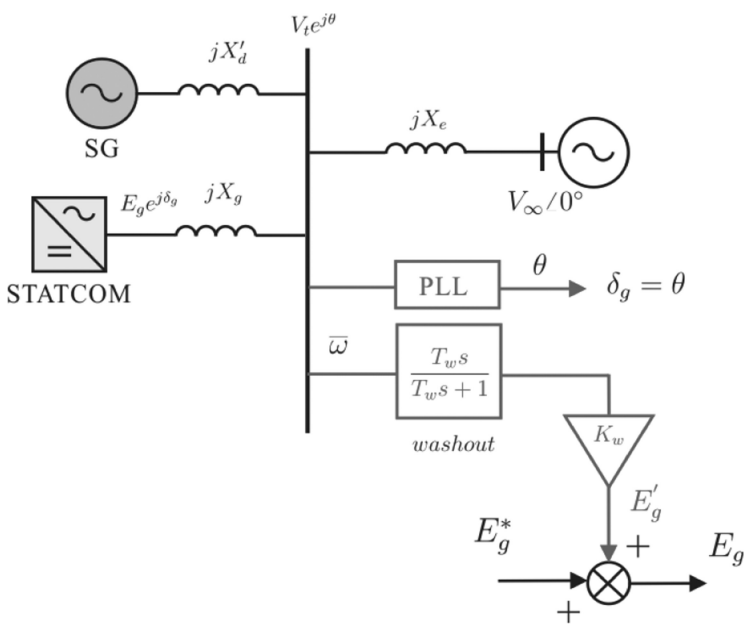


FIGURE 8.8 Block diagram of a STATCOM stabilizer.

Considering that $\delta_g = \theta$ and according to equation (8.16), the reactive power exchange by the STATCOM only depends on voltage E_g with respect to bus voltage V_t.

$$Q_g = \frac{V_t}{X_g}\left(E_g - V_t\right) \tag{8.27}$$

From the frequency measured in the bus, $\bar{\omega}$, and using the same washout filter, a compensation voltage E'_g is obtained as follows

$$E'_g = K_w\left(\frac{T_w s}{T_w s + 1}\right)\bar{\omega} \tag{8.28}$$

As shown in Figure 8.8, the signal E'_g is added to the internal voltage reference E^*_g. Unlike to the grid-forming stabilizer, in this case, the LFO damping is achieved by acting on the voltage instead of the frequency. In this case, the compensator is denoted as POD-Q since the voltage variation leads to reactive power exchange.

Linearizing equation (8.28), $\Delta E'_g$ is now different to zero and equal to

$$\Delta E_g = K_w\left(\Delta\bar{\omega} - \Delta z\right) \tag{8.29}$$

where Δz is the same auxiliary variable used in equation (8.22).

Now, a fifth-order model completes the set of dynamic equations. Only one additional equation, corresponding to the washout filter, is considered equation (8.26). All dynamic equations of the SG and the four algebraic equations of the previous case are the same. In this case, matrices A, B, C, and D are different since now $\Delta E'_g$ is not null and depends on $\Delta\bar{\omega}$ and Δz. The expression of the matrices of the linearized model is also given in Appendix B.

8.5 STABILITY ANALYSIS

8.5.1 Small-Signal Stability

The aim of this section is to analyze the small-signal stability of the linearized systems described previously. An eigenvalues analysis will be carried out in order to evaluate the impact on the oscillation modes when using the stabilizers associated to grid-forming converters and STATCOM. All parameters used in the model are shown in Table 8.3 of Annex A.

System eigenvalues are obtained from the state matrix which is calculated using the parameters of Table 8.3 and the state and algebraic variables evaluated at the operation point. Table 8.4 of Annex A gives these state and algebraic variables in an operation point corresponding to a voltage at the generator bus with module $V_t = 1$ pu and angle $\theta = 9^0$.

Figure 8.9 shows, in blue, the eigenvalues of a single-machine infinite-bus case corresponding to the electromechanical and the flux-decay oscillation modes of an SG. These modes are also represented when using the stabilizers POD-P (grid-forming) and POD-Q (STATCOM). In the case of the grid forming, the oscillation modes are represented in red, and for the STATCOM, the oscillation modes are represented in green.

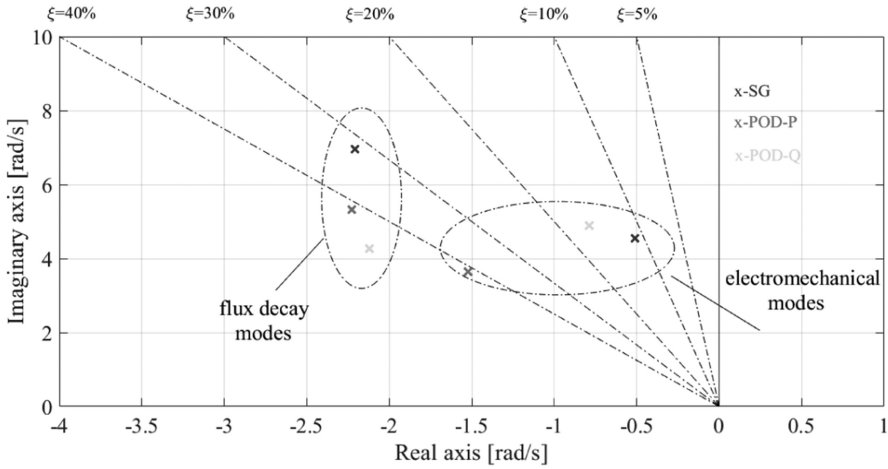


FIGURE 8.9 Eigenvalue loci for a single-machine infinite-bus system.

This figure shows that the electromechanical oscillation mode is significantly damped up to a damping factor of 40.2% when a POD-P is used if compared to the baseline case defined as an SG without any PEC stabilizer. In the case of using a POD-Q with a STATCOM, the electromechanical oscillation mode is only damped up to 15.84%, which demonstrates that it is not so effective as the POD-P stabilizer. However, a POD-Q is very effective to damp the oscillation modes associated to the flux-decay since a POD-Q acts directly on the voltage. The damping factor of this mode is significantly high, 44.47%, in the case of the POD-P.

In Table 8.1 system eigenvalues, damping factor and natural frequency, as well as the corresponding participation factors for each state variable, are shown considering the three cases analyzed. In the baseline case, eigenvalues λ_{12}, corresponding to the electromechanical mode, are dominated by the states $\Delta\bar{\omega}$ and $\Delta\delta$ with a participation factor of 0.38 each one. While for the flux-decay mode λ_{34}, $\Delta E'_q$ and ΔE_{fd} are their dominant states with a participation factor of 0.41 and 0.39, respectively.

The eigenvalue analysis, when a POD-P stabilizer is used, shows that participation factors of electromechanical and flux-decay modes are more distributed between states $\Delta E'_q$, $\Delta\bar{\omega}$, $\Delta\delta$, and ΔE_{fd}. Besides, in this case, two additional non-oscillatory stable eigenvalues, λ_5 and λ_6, are observed.

The eigenvalue λ_5 is dominated by the state $\Delta\delta_g$, and λ_6 by the auxiliary state Δz. In the first case, the participation factor of the dominant state is 0.95 with natural frequency of 6.74 Hz, and for the second one, the participation factor of Δz is 0.96 with a natural frequency of 0.16 Hz. These two eigenvalues have not been represented in Figure 8.9.

A similar study has been carried out for a POD-Q stabilizer. The electromechanical modes are slightly modified by the STATCOM action, and the participation factor of the dominant states is very similar to the baseline case. However, for the flux-decay mode, states $\Delta E'_q$ and ΔE_{fd} are no longer so dominant and a higher

TABLE 8.1
Eigenvalues, Damping Factor and Natural Frequency, as well as the Participation Factors for the Three Cases Analyzed

Eigenvalues	ξ (%)	ω_n (Hz)	$\Delta E'_q$	$\Delta\delta$	$\Delta\bar{\omega}$	ΔE_{fd}	$\Delta\delta_g$	Δz
			SG (baseline case)					
$\lambda_{12} = -0.51 \pm j4.55$	11.17	0.72	0.14	0.38	0.38	0.10	0.00	0.00
$\lambda_{34} = -2.21 \pm j6.95$	30.25	1.10	0.41	0.10	0.10	0.39	0.00	0.00
GFC (POD-P)								
$\lambda_{12} = -1.52 \pm j3.63$	40.20	0.58	0.32	0.25	0.17	0.10	0.01	0.02
$\lambda_{34} = -2.22 \pm j5.33$	40.30	0.85	0.24	0.27	0.24	0.20	0.03	0.02
$\lambda_5 = -42.38$	100	6.74	0.00	0.00	0.05	0.00	0.95	0.00
$\lambda_6 = -1.03$	100	0.16	0.00	0.04	0.00	0.00	0.00	0.96
			STATCOM (POD-Q)					
$\lambda_{12} = -0.78 \pm j4.90$	15.84	0.78	0.19	0.37	0.31	0.12	0.00	0.01
$\lambda_{34} = -2.12 \pm j4.27$	44.47	0.68	0.39	0.17	0.10	0.32	0.00	0.02
$\lambda_5 = -0.95$	100	0.15	0.00	0.04	0.00	0.00	0.00	0.96

contribution of $\Delta\delta$ is observed. A non-oscillatory mode λ_5, corresponding to the washout filter, appears in this case. Its natural frequency is 0.15 Hz and has not been represented in Figure 8.9 either.

8.5.2 Time-Domain Response

In this section, the time-domain response of active and reactive power transmitted through a line in a single-machine infinite-bus configuration is analyzed. At $t = 0$, a step change of 0.5 pu is applied to the generator torque reference. Figure 8.10 shows the active power response for the three cases considered.

As shown in Figure 8.10, the POD-P stabilizer damps the active power oscillation in just 3 s with only two oscillations before reaching steady state. On the contrary, POD-Q damps the active power oscillation very slightly in comparison to the baseline case. In this case, the steady state is reached in about 10 s.

Figure 8.11 shows a similar response corresponding to the reactive power. The response times and oscillation modes are consistent with the eigenvalue analysis performed in the previous subsections.

8.6 TWO-AREA SYSTEM

In this section, the damping of the inter-area oscillations is addressed by applying the control schemes proposed above. Inter-area oscillations usually appear when two synchronous areas are interconnected through a weak tie-line.

Figure 8.12 shows a single line diagram where two synchronous areas, represented by SG1 and SG2, are interconnected through a double interconnection line.

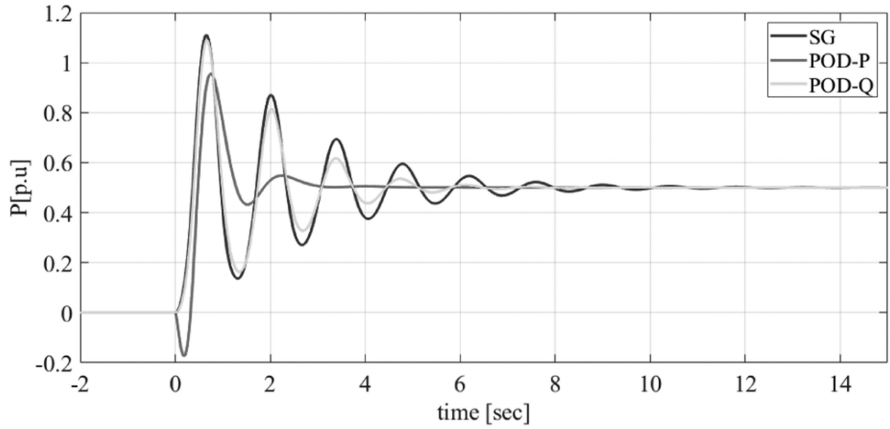


FIGURE 8.10 Active power response to a torque step change of 0.5 pu in the SG.

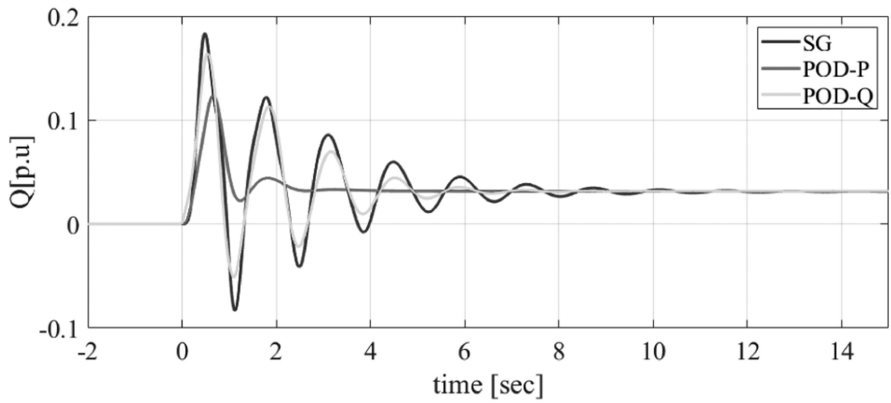


FIGURE 8.11 Reactive power response to a torque step change of 0.5 pu in the SG.

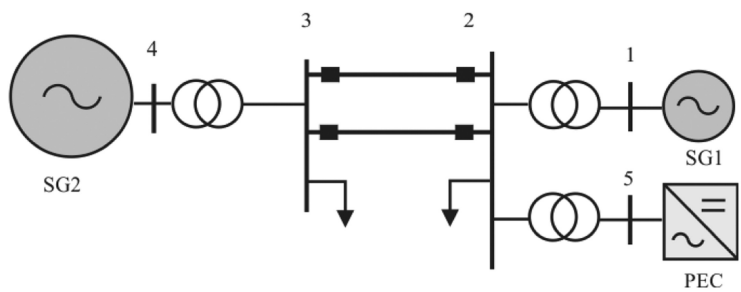


FIGURE 8.12 Single line diagram of two-area system with a PEC.

The equivalent generator SG1 has a power rating of 1,500 MVA, and it is connected to the bus 1 through a step-up transformer. At the bus 1 is also connected a PEC with the same power rating than SG1, 1,500 MVA, using another transformer. Likewise, the equivalent generator SG2 is connected to the other end of the line and has a power rating of 4,000 MVA. In Table 8.2, it is shown the steady-state operation point of the two-area system before applying a load change. Each line between buses 2 and 3 transmits 50 MW, being the active and reactive power injected by the PEC equal to zero.

In this study, a variable reactance, X_L, corresponding to each transmission line, has been considered. Its value varies from 0.01 to 0.9 pu, considering a power rating of 100 MVA and a rated voltage of 400 kV. The step-up transformer of SG2 has a short-circuit impedance of 0.3%, and 1% for the SG1 transformer. In both cases, a 100 MVA power base has been considered. The loads have been modeled as ZI loads, considering constant current when modeling active power changes and constant impedance for reactive power changes.

SG1 and SG2 have been modeled using the well-known GENROU model. The steam turbine and the speed governor have been modeled using the IEEEG1 model. And for the excitation system, a model IEEE-ST1 has been used. In Appendix C, all the parameters of these models are given.

8.6.1 Eigenvalues Loci Under Variation of Parameters

Figure 8.13 shows the oscillation modes of the two-area system in the three cases previously analyzed and when the reactance X_L of the transmission lines varies between 0.01 and 0.9 pu First, the electromechanical and flux-decay oscillation modes of generators SG1 and SG2 are shown when no PEC is connected. Subsequently, these oscillation modes are represented when a PEC is connected to bus 2, first acting as a grid forming using a POD-P stabilizer, or after as STATCOM applying a POD-Q control.

Small-signal stability analysis of power systems can be studied using different methodologies. Among them is worth to mention the frequency-domain methods, based on the impedance model, and phase-amplitude dynamics model, or the time-domain methods based on the state-space model [10]. This last one has been used in this section. Besides, as in the previous sections, the small-signal stability analysis

TABLE 8.2
Steady-State Operation of Generators and Loads

Generators	MW	Vt
BUS1	1350	1.0
BUS2	3900	1.0
BUS3	0.0	1.0
LOADS	MW	MVar
BUS2	1250	0.0
BUS3	4000	0.0

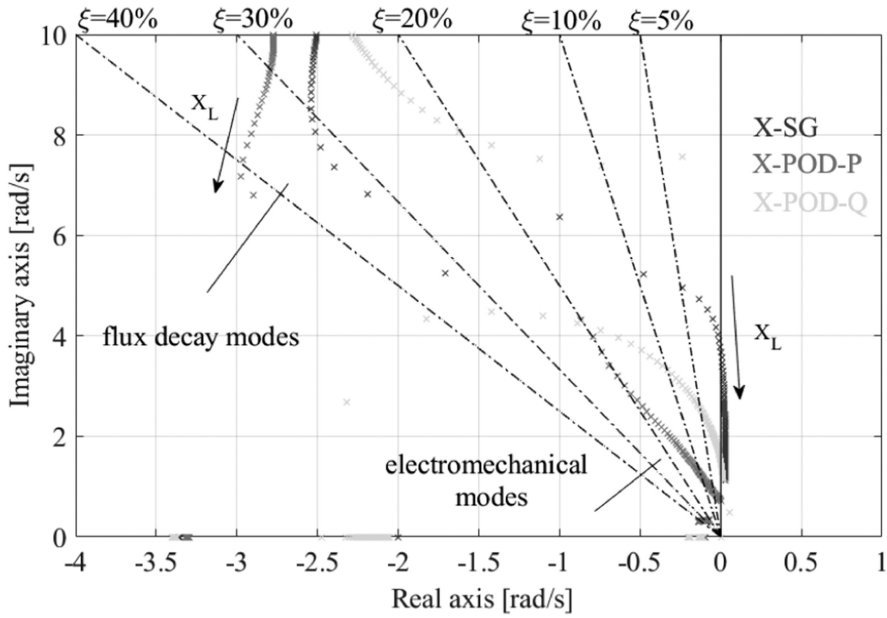


FIGURE 8.13 Eigenvalue loci of the two-area system under X_L variation.

corresponding to the two-area system could be addressed analytically; however, a numerical method has been used in order to avoid making the exposition overly cumbersome.

The eigenvalue loci corresponding to LFO of the two-area system is shown in Figure 8.13. In this figure, the three cases analyzed are represented when X_L varies from 0.01 to 0.9 pu. As shown in Figure 8.13, when the interconnection between areas is weaker (higher X_L value), the natural frequency of the mechanical oscillation modes (inter-area modes) is reduced, as well as the damping factor. If a stabilizer is not applied, the system could become unstable when the line has a reactance higher than 0.13 pu.

When a POD-P stabilizer is used on the PEC, acting as a grid-forming converter, the damping factor corresponding to the electromechanical modes increases up 20% regardless of X_L. As for the flux-decay oscillation modes, the same damping effect is also observed. However, in the case of using a POD-Q, damping of the flux-decay oscillation modes is reduced when X_L increases. In conclusion, POD-P is more effective than POD-Q in all the cases analyzed.

8.6.2 Dynamic Response to a Load Change

In this section, the dynamic response of the two-area system is analyzed when a power reduction of 100 MW is applied to the load connected in bus 2 at $t = 0$. The initial conditions, before the load change, are those shown in Table 8.2. The reactance chosen for the transmission line is $X_L = 0.1$ pu (0.2 pu in each line) since gives rise to a stable operation even in case that no stabilizer is used.

Figure 8.14 shows the active and reactive power transmitted through one of the transmission lines after a load change. The dynamic response of both powers is represented in the three cases analyzed. In the first case, when the PEC is disabled (SG case), power oscillations show a frequency of 0.64 Hz and are weakly damped. In the second case, corresponding to the POD-P stabilizer, LFO is reduced reaching a damping factor of 20%. In case of using a POD-Q stabilizer, the power oscillations are also reduced in comparison to the first case; however, it is less effective than the POD-P. Initially, each interconnection line transmits 50 MW and 5 MVar. After the load change, the power transmitted is 100 MW and 35 MVar in steady state.

Figure 8.15 shows the active and reactive power generated by SG1. Active power remains constant in steady state after the load change (1,350 MW) and the reactive power increases slightly. The reactive power presents poorly damped oscillations

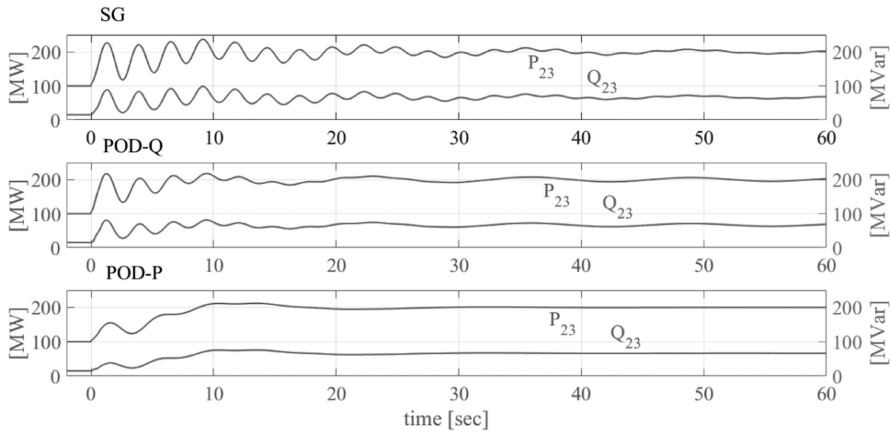


FIGURE 8.14 Active and reactive power transmitted by the two transmission lines under a load change.

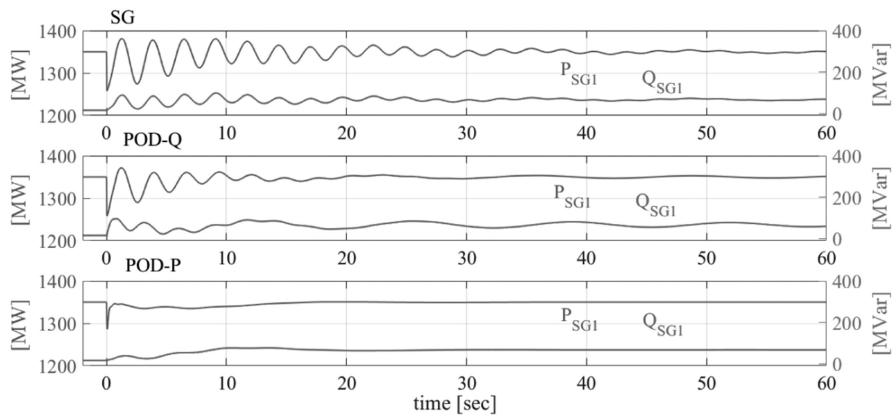


FIGURE 8.15 Active and reactive power response of SG1.

when a POD-Q is used, which is in accordance with the eigenvalue analysis shown in Figure 8.13.

As before, Figure 8.16 shows the active and reactive power generated by SG2. In this case, the active power is reduced by 100 MW (from 3,900 to 3,800 MW) in such a way that the load at node 3 (4,000 MW) is covered with 200 MW from the transmission lines and 3,800 MW from SG2.

Figure 8.17 shows the active and reactive power exchange of the PEC when a POD-P or a POD-Q stabilizer is used. In the first case, the grid-forming converter only exchanges active power, being reactive power equal to zero. After the load change, the GFC absorbs a maximum power of 70 MW and generates a maximum power of 30 MW, and stop exchanging active power when the frequency reaches steady state. In the STATCOM case, its behavior is just the opposite, it only exchanges reactive power being the active power exchange equal to zero. The reactive power exchange varies by ±25 MVar.

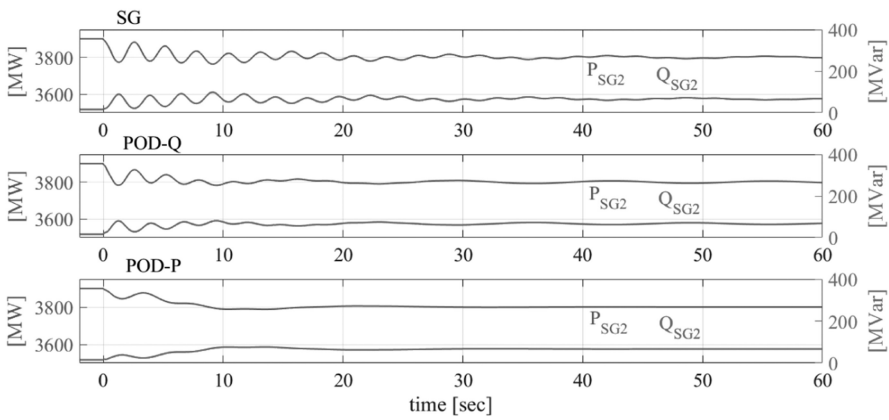


FIGURE 8.16 Active and reactive power response of SG2.

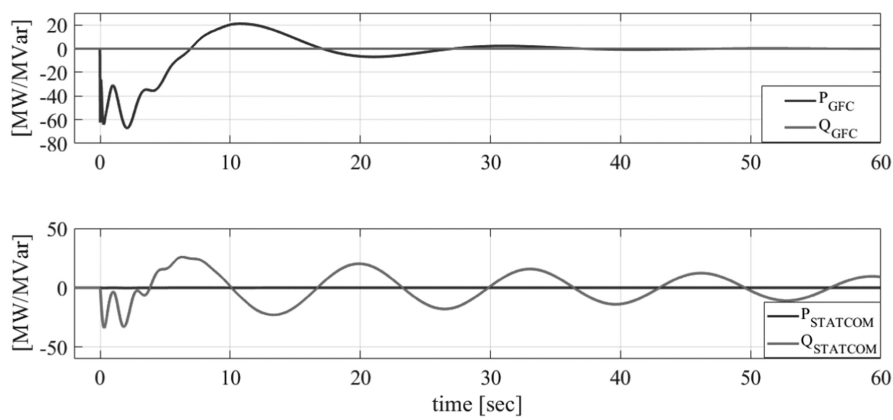


FIGURE 8.17 Active and reactive power exchange by a GFC and a STATCOM.

In conclusion, the POD-P stabilizer is more effective than the POD-Q to damp power oscillations. However, the grid-forming converters that use POD-P stabilizers require an energy buffer connected to the DC bus in order to exchange active power. While the STATCOM does not require any energy storage, but its damping effectiveness is much more limited. Anyway, a grid-forming converter with a POD-P stabilizer does not require a battery with a high energy capacity. According to the example of this section, power damping of a line transmitting 100 MW can be achieved through a net energy absorption of approximately 40 kWh that for a line of these characteristics is not an excessively high value.

8.6.3 Dynamic Response to a Line Tripping

This section shows the dynamic response of the two-area system when one of the two lines of the interconnection is tripped (see Figure 8.12). Before the tripping, each line transmitted 50 MW, but when the disturbance appears (at $t = 0$ s), the active power oscillates until reaching 100 MW in steady state, and it was the total power transmitted by the two lines before the tripping. Figure 8.18 shows the power oscillation through the healthy line after the tripping, in the cases of no compensation (top graph) and when using the POD-P and POD-Q stabilizers (bottom graph). In last two cases, power oscillation damping is achieved in less than 4 s.

During the line tripping event, the active and reactive power exchanges by the grid-forming and the STATCOM are shown in Figure 8.19. At the top, it is observed that the active power of the grid-forming converter varies between ±40 MW during the first oscillation. Likewise, at the bottom, corresponding to the STATCOM, the reactive power oscillates between ±20 MVar. In this case, it can be concluded that the STATCOM is more effective since it exchanges less power than the grid-forming converter while achieving nearly the same damping.

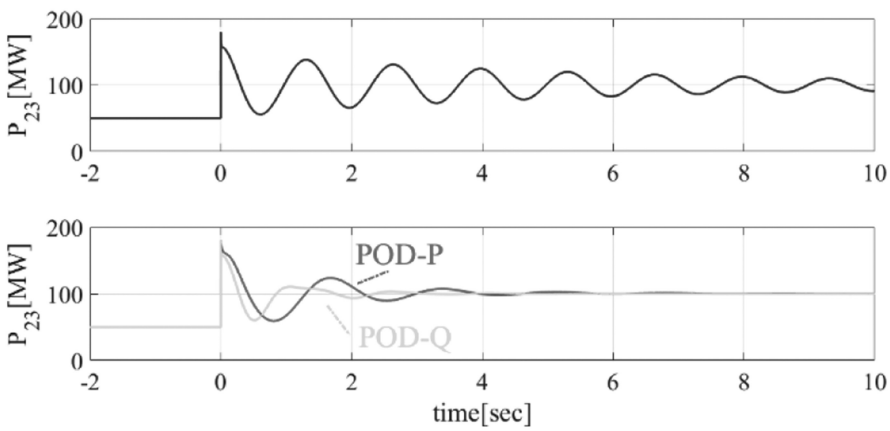


FIGURE 8.18 Active power transmitted by the interconnection during a line tripping.

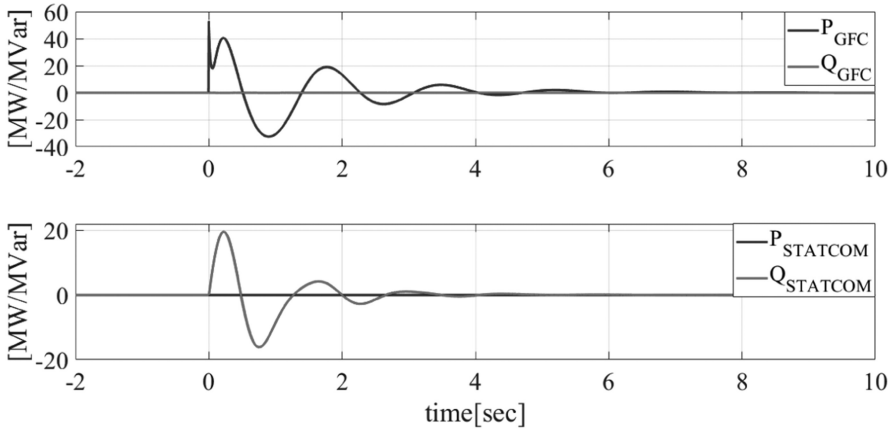


FIGURE 8.19 Active and reactive power exchange by the GFC and the STATCOM.

8.7 CONCLUSIONS

Nowadays, there is a growing concern about power system stability as a consequence of the increasing incorporation of renewable generation. Among stability issues, low-frequency oscillations arise as one of the most important concerns. Traditionally, these oscillations have been dampened by tuning the PSS of conventional SGs. However, modern PECs can contribute to provide this service efficiently. In this chapter, two stabilizers have been proposed; demonstrating the ability of both stabilizers to damp low-frequency oscillations in power systems. The first stabilizer, named POD-P, has been implemented in a grid-forming converter acting on the power angle, and consequently, exchanging active power to damp power oscillations. The second stabilizer, named POD-Q, has been implemented in a STATCOM using the internal voltage magnitude to damp system oscillations and exchanging only reactive power. Small-signal stability analysis has been carried out based on the linearized model of the system demonstrating the effectiveness of the proposed PODs. This effectiveness has also been proved through simulation, using comprehensive models of the nonlinear system. Simulation results, as well as the stability study, show the superiority of the POD-P stabilizer over the POD-Q, but at a cost of having to use some kind of energy supply in the DC bus to support the power interchange during system stabilization.

APPENDIX A. SYSTEM PARAMETERS AND OPERATION POINTS

TABLE 8.3
System Parameters

Parameter	Symbol	Value	Units
Stator resistance	R_s	0.0	pu
d-axis synchronous reactance	X_d	1.8	pu
q-axis synchronous reactance	X_q	1.7	pu
d-axis transient reactance	X'_d	0.3	pu
Line resistance	R_e	0.0	pu
Line reactance	X_e	0.3	pu
d-axis transient open-circuit constant	T'_{d0}	8.0	s
Inertia constant	H	6.175	s
Damping factor	D	0.0	pu
Exciter time constant	T_A	0.2	S
Exciter gain	K_A	200	pu
Frequency (50 Hz)	ω_s	314.16	rad/s
PEC reactance	X_g	0.15	pu
Washout time constant	T_w	1.00	S
Washout gain	K_w	3.00	pu
Droop gain	R_g	0.05	pu

TABLE 8.4
State Variables and Algebraic Variables at the Operation Point when $V_\infty = 1$ pu, $V_t = 1$ pu, and $\theta = 9^0$

State Variables	Symbol	Value	Units
q-axis transient voltage	E'_q	0.8793	pu
q-axis angle	δ	48.647	degrees
Rotor speed	ω	314.159	rad/s
Excitation voltage	E_{fd}	1.4258	pu
Algebraic Variables	Symbol	Value	Units
d-axis current	I_d	0.3643	pu
q-axis current	I_q	0.3753	pu
d-axis voltage	V_d	0.6381	pu
q-axis voltage	V_q	0.7700	pu
Inputs	Symbol	Value	Units
Mechanical torque	T_m	0.5214	pu
Reference voltage	V_{ref}	1.0071	Pu